

TATA TECHNOLOGIES LTD.

TECH PULSE FY-26 | AI/ML Capstone Project

PROJECT SYNOPSIS

Student Details

Institute/University	Pimpri Chinchwad College of Engineering (PCCOE), Pune
Programme / Batch	B.E. Computer Engineering, Final Year
Student Name	Sajjan Pawar
PRN	124B2B037

1. Case Study ID and Project Title

Case Study ID (CS1 to CS5)	CS3 (RAG-based Document Q&A; System)
Project Title / Topic	AUTOSAR HLD Document Analysis Assistant using RAG with Citations

2. Objectives

1. Build a Retrieval-Augmented Generation (RAG) assistant that answers questions about AUTOSAR High-Level Design (HLD) documents with page citations.
2. Implement safe refusal mechanism to answer 'Not found in the document' when the answer is not present.
3. Measure retrieval accuracy, answer accuracy, citation accuracy, and correct refusal rate on 30 ground-truth questions.
4. Demonstrate grounding and citation capabilities as a Responsible AI measure.

3. Scope

In Scope – To Be Implemented

- PDF text extraction using pdfplumber, section-wise chunking (800 chars), ChromaDB vector store
- Semantic retrieval using BAAI/bge-small-en-v1.5 embeddings, top-4 retrieval, distance threshold 0.335
- Local LLM (llama3.2:3b via Ollama) with strict citation prompt requiring [Page N] citations
- Streamlit web UI for interactive Q&A; with retrieved evidence display
- Evaluation on 30-question test set with 4 metrics
- Synthetic 14-page BCM HLD document (no real company data)

Out of Scope – Will Not Be Implemented

- Multi-document knowledge base • OCR for scanned PDFs • Diagram understanding • Revision comparison
- Graph-based relationships • Role-based access control • Production deployment (FastAPI, Docker, PostgreSQL)

4. Proposed Approach

Component	Choice
Pre-trained model	llama3.2:3b via Ollama (local), Temperature 0.0
Embedding model	BAAI/bge-small-en-v1.5 (384-dim, Hugging Face, offline)
Vector store	ChromaDB v1.5.9, persistent, cosine distance, top-4
RAG pipeline	Query → BGE embed → ChromaDB retrieve → threshold check (0.335) → LLM generation + [Page N]

Service layer and UI	Streamlit 1.65.0 with example questions, LLM toggle, retrieved evidence display
Tools / rules	Distance refusal, citation extraction, keyword evaluation, section-aware chunking

5. Input Data / Knowledge Base

Data Sources	Synthetic 14-page BCM HLD (BCM_HLD_v1.0.pdf) created with ReportLab
Size and Format	14 pages, ~18 KB, PDF with text and tables (no diagrams)
Preprocessing	pdfplumber extraction → section identification → 800-char chunks → 14 chunks indexed
Test Set	30 questions (28 answerable, 2 unanswerable) with ground truth keywords and pages

6. Expected Outcomes and Evaluation Plan

Expected Outputs: Working Streamlit app, evaluation results (CSV, JSON, markdown), technical report

Evaluation Metrics:

- Retrieval hit rate: % with expected page in top-4
- Answer accuracy: % with all required keywords
- Citation accuracy: % with expected page cited
- Correct refusal rate: % unanswerable questions refused

Evaluation Mode: Extractive mode (--no-llm) because Ollama was not available. LLM generation code implemented in rag.py but not evaluated.

Responsible AI Measures:

- Grounding: Mandatory [Page N] citations, retrieved evidence shown in UI
- Privacy: Synthetic data only, local execution, no data collection
- Security: No cloud APIs, ChromaDB on localhost
- Human oversight: UI disclaimer, verification mechanism, limitations documented

7. Planned AI-Tool Usage

AI Tool / Model	Purpose	Artifact Affected
GitHub Copilot	Code autocompletion	Code/*.py
ChatGPT-4	Debugging assistance	ingest.py, rag.py
Claude Sonnet (Kiro)	Technical report drafting	Documentation, README

All code and documentation reviewed, tested, and understood by student.

Student Declaration

- I will not change the approved case-study scope without prior faculty approval.
- I confirm this work represents my own understanding and contribution.

Name and Signature: Sajjan Pawar _____ Date: 07/10/2026